Soft Skills, Hard Times:
Why New College Graduates Are Falling Behind

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October 27, 2026

The views expressed should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or any other person associated with the Federal Reserve System.

Motivation

- The unemployment rate of young college graduates has worsened since 2011
  - 18% **lower** than average in 2011
  - 33% **more** than average in 2024

- Leading explanations:
  - Artificial Intelligence (AI), remote work
  - But this trend predates the AI boom and remote work expansion

- This paper:
  - Rising demand for soft skills (Deming, 2017; Edin et al., 2022)
  - Explains one-fifth of the pre-COVID rise in young workers' unemployment

Overview

- **Empirical evidence on soft skills**
  - Young-old wage gap higher in soft skill-intensive jobs
  - Demand for soft skills is rising
- **Search-and-matching model**
  - Heterogeneous workers and firms
  - On-the-job skill accumulation
- **Parametrization**
  - Match model to US labor market moments (1951-2012)
  - Simulate the observed rise in skill demand (2012-2024)

Why Rising Soft Skill Demand Worsens Youth Unemployment

- **Young College Graduates (Labor Supply):**
  - Soft skills are accumulated on the job
  - Young workers unable to acquire them before they enter labor market
- **Employers (Labor Demand):**
  - Rising soft skill demand (Deming, 2017; Edin et al., 2022)
  - Vacancy distribution shifts toward higher-skilled workers
- **Consequence:**
  - Rising gap between skill demand and skill supply
  - Unemployment rate of young college graduates rises

Roadmap

1. Empirical Evidence
2. Model
3. Model Parametrization
4. Counterfactual Exercise
5. Future Work

Roadmap

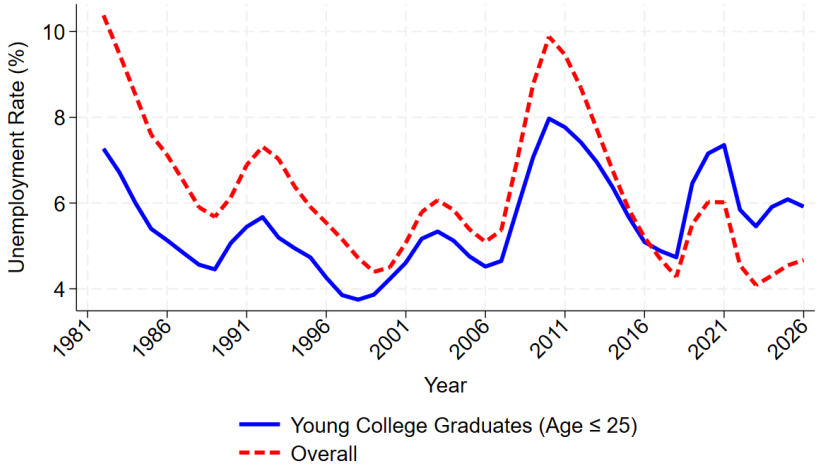
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Data Sources

1. **Lightcast (formerly known as Burning Glass)**
   - Near universe of online job postings in the US
   - Time period: 2011–2024
   - 229 million job postings
   - 34,351 skills

2. **Current Population Survey (CPS)**
   - Monthly survey data on labor market outcomes
   - Focus on college graduates
   - Young $= 18 - 25$ years old
   - Old $= 26 - 55$ years old

The Unemployment Rate Of Young Workers Is Rising



Non-College Workers

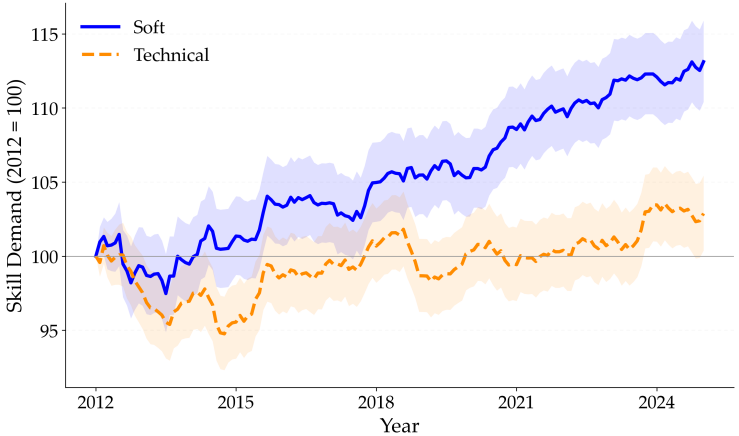
Soft Skills Definition

"... [I]nterpersonal and thinking skills needed to interact successfully with people and to perform efficiently and effectively in the workplace." —*O*NET*

- Skills that do not satisfy the above criteria are considered technical skills.
- Manually classify top 1000 skills by share in Lightcast.
- Classify remaining skills using AI.
- Results are robust even if entire classification done using AI.

Common Skills | Example: Technical | Example: Soft

Rising Demand For Soft Skills



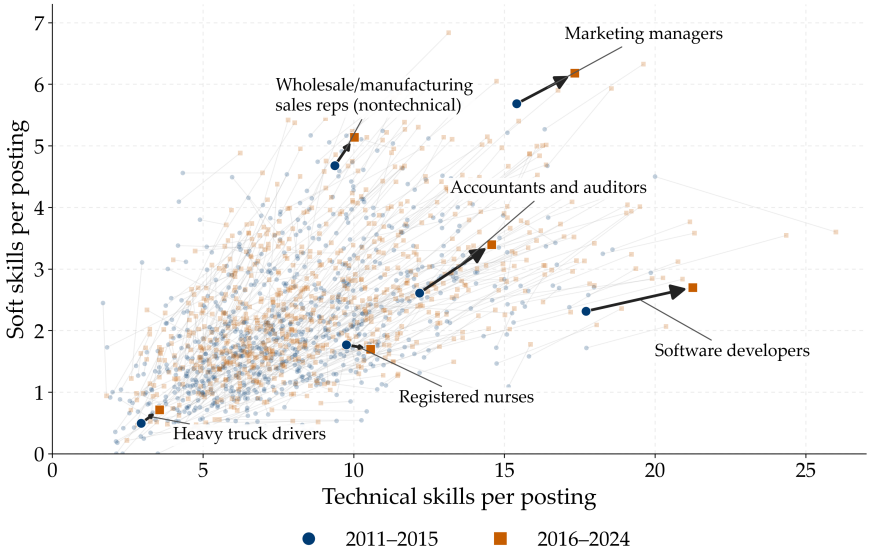
Consistent with Deming (2017) and Edin et al. (2022)

Rising Demand For Soft Skills

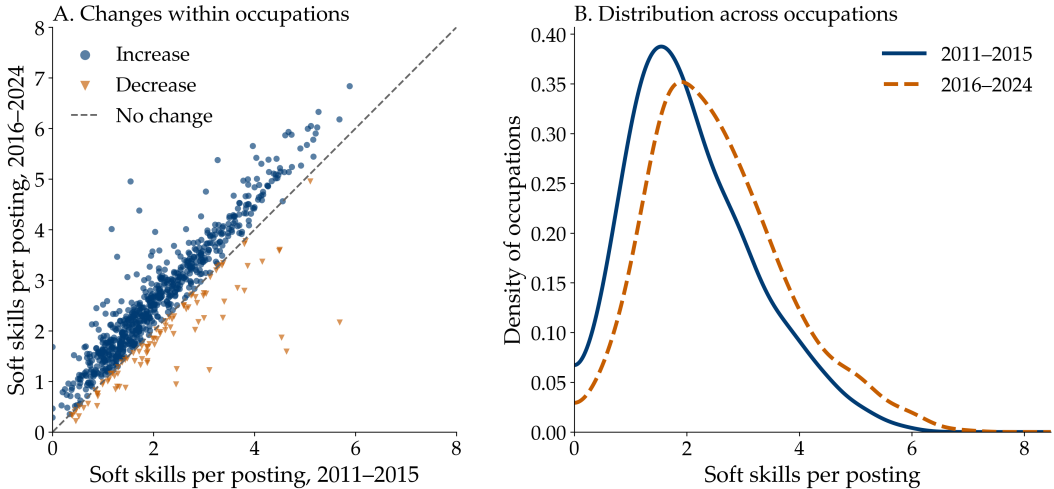
Dependent Variable:	Log(All Skills)	Log(Soft Skills)	Log(Technical Skills)
% change per year	0.60***	1.20***	0.48***
	(0.12)	(0.24)	(0.12)
Fixed-Effects:			
Occupation	Yes	Yes	Yes
Month	Yes	Yes	Yes

- **Skill demand measure:** Average number of skills per posting
- Soft skill demand increases by **15.6%** from 2012 to 2024
- Technical skill demand increases by **6.2%**

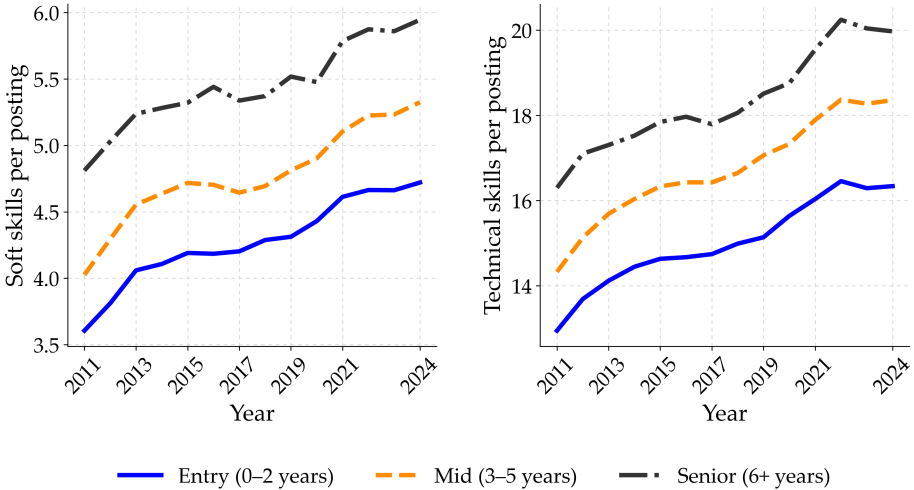
Skill Demand Is Rising Within Most Occupations



Soft Skill Demand Is Rising Within Most Occupations



Skill Demand Is Rising Across Work Experience Levels



How Well Do Young Workers Perform in Skill-Intensive Jobs?

Construction

- **Wages:** CPS MORG, 2011–2024, wage-employed, age ≤ 55
- **Skill demand:** Lightcast average soft / technical skills per posting
- Merged at the **occupation** $\times$ **month** level
- **Young** $=$ age ≤ 25; **Old** $=$ age 26–55

Specification

$$\begin{aligned}\log w_{iot} = {} & \beta_s \text{Soft}_{ot} + \beta_\tau \text{Tech}_{ot} + \gamma\, \text{Young}_i \\ & + \delta_s (\text{Young}_i \times \text{Soft}_{ot}) + \delta_\tau (\text{Young}_i \times \text{Tech}_{ot}) \\ & + X_i' \Theta + \lambda_t + \varepsilon_{iot}\end{aligned}$$

- X_i: sex, race, education; λ_t: month FE

Young Workers Earn Less Than Older Workers

Dependent Variable: Log(Wages)	College Educated	Non-College Educated
Young (γ)	-0.3724^{***} (0.0396)	-0.2185^{***} (0.0250)
Soft skills (β_s)	0.0266^{*} (0.0155)	0.0072 (0.0113)
Technical skills (β_τ)	0.0296^{***} (0.0031)	0.0355^{***} (0.0045)
Young $\times$ Soft (δ_s)	-0.0300^{**} (0.0145)	-0.0252^{***} (0.0095)
Young $\times$ Technical (δ_τ)	0.0093^{***} (0.0033)	-0.0080^{***} (0.0027)
Observations	572,091	931,070

Soft and Technical Skills Are Valued In The Labor Market

Dependent Variable: Log(Wages)	College Educated	Non-College Educated
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Young College Graduates Have Technical Skills, Lack Soft Skills

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Observations	572,091	931,070

- Consistent with an environment where:
  - Soft skills can only be accumulated on the job ($\delta_s < 0$ everywhere)
  - Technical skills are acquired in college ($\delta_\tau > 0$ for college educated, $\delta_\tau < 0$ for non-college educated)

Coefficients Over Time

Roadmap

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Model Agents

	Firms	**Workers**
Type	$y \in [0,1]$ (technology)	$x \in [0,1]$ (skills)
Actions	Post vacancies $v(y)$	Accept or reject job offers
Value if matched	$P_t(x,y,z) - W_t(x,y,z)$	$W_t(x,y,z)$
Value if unmatched	0	$B_t(x)$

Match Creation and Destruction

- Total meetings created: $M_t = M(L_t, V_t) = \min[\alpha_M L_t^{\omega} V_t^{1-\omega}, L_t, V_t]$
  - where V_t is the total vacancies created
  - and L_t is the total number of job searchers

- $\lambda_{w,t} = M_t/L_t$ is the job finding rate for workers
- $\lambda_{f,t} = M_t/V_t$ is the job filling rate for firms
- $\theta_t = V_t/L_t$ is the labor market tightness

- Matches are destroyed at the rate δ every period

Production Function And Outside Options

- Production function (Lise and Robin, 2017):

$$p_t(x, y, z) = z_t(p_1 + p_2 x + p_3 y + p_4 x^2 + p_5 y^2 + p_6 xy)$$

- Flow value of home production (Hall, 2005):

$$b_t(x) = 0.7 \times p_t(x, y^*(x, 1), 1)$$

where

$$y^*(x, 1) = \arg\max_y p_t(x, y, 1)$$

Values and Surplus: Final Expressions

- Value from unemployment: $B_t(x) = b(x) + \beta\, \mathbb{E}_t B_{t+1}(x)$
- Present value of a match:

$$P_t(x,y,z) = p(x,y,z_t) + \beta\, \mathbb{E}_t \Big[\underbrace{\big(\mathbf{1}\{P_{t+1}(x,y,z) < B_{t+1}(x)\}}_{\text{Endogenous separations}} + \underbrace{\delta \mathbf{1}\{P_{t+1}(x,y,z) \geq B_{t+1}(x)\}\big)}_{\text{Exogenous separations}} B_{t+1}(x) + \underbrace{(1-\delta)\mathbf{1}\{P_{t+1}(x,y,z) \geq B_{t+1}(x)\}}_{\text{If the match survives}} P_{t+1}(x,y,z) \Big]$$

- Surplus from a match: $S_t(x,y,z) \equiv P_t(x,y,z) - B_t(x)$

With Skill Accumulation

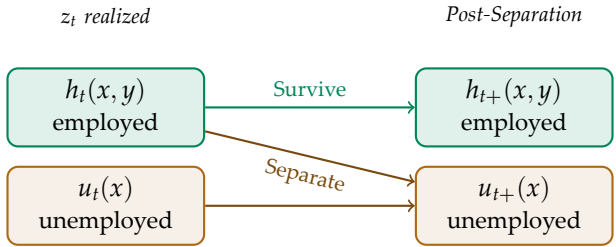
Timeline: Beginning of Period t

$h_t(x, y)$
employed

$u_t(x)$
unemployed

$$\underbrace{u_t(x)}_{\text{unemployed}} + \int \underbrace{h_t(x, y)}_{\text{employed at firm y}} \, \mathrm{d}y = \ell(x)$$

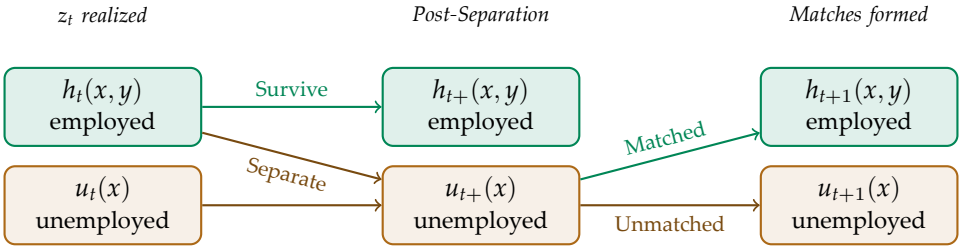
Timeline: Separations



$$u_{t+}(x) = u_t(x) + \int \Big[\underbrace{\mathbf{1}\{S_t(x,y,z) < 0\}}_{\text{endogenous separation}} + \underbrace{\delta\, \mathbf{1}\{S_t(x,y,z) \geq 0\}}_{\text{exogenous separation}} \Big] h_t(x,y)\, \mathrm{d}y$$

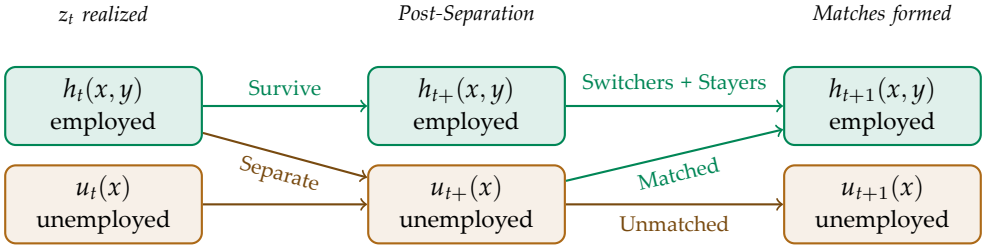
$$h_{t+}(x,y) = \underbrace{(1-\delta)\, \mathbf{1}\{S_t(x,y,z) \geq 0\}}_{\text{no separation}}\, h_t(x,y)$$

Timeline: Matches from Unemployment



$$u_{t+1}(x) = u_{t+}(x) - u_{t+}(x) \int \underbrace{\lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x,y,z) \geq 0\}}_{\text{P(Accepting a job at firm y)}} \, \mathrm{d}y$$

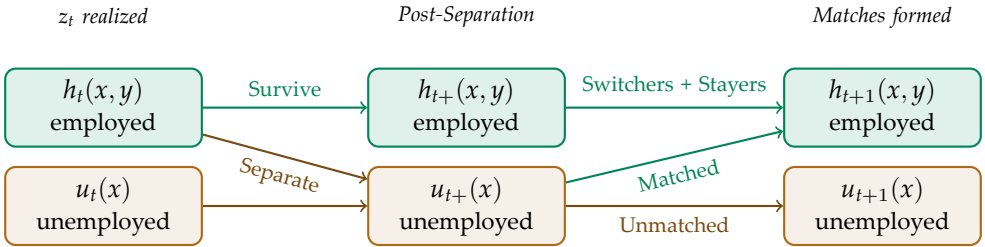
Timeline: Matches from Employment



$$h_{t+1}(x,y) = h_{t+}(x,y) - \underbrace{h_{t+}(x,y) \int s\lambda_{w,t} \frac{v_t(y')}{V_t} \mathbf{1}\{S_t(x,y') > S_t(x,y)\}\, \mathrm{d}y'}_{\text{Poached by other firms}}$$

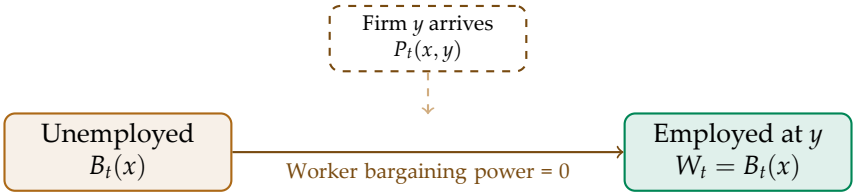
$$+ \underbrace{\int h_{t+}(x,y')\, s\lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x,y) > S_t(x,y')\}\, \mathrm{d}y'}_{\text{Poached from other firms}} + \underbrace{u_{t+}(x)\, \lambda_{w,t} \frac{v_t(y)}{V_t} \mathbf{1}\{S_t(x,y) \geq 0\}}_{\text{hires from unemployment}}$$

Timeline: After Matches Are Formed

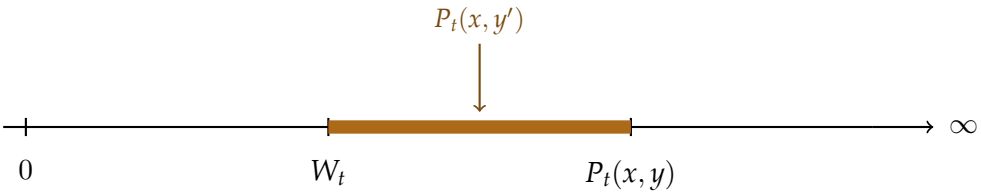


- Production takes place
- Employed workers accumulate skills: $x_{t+1} = x_t + \Delta$ with probability π, $x_{t+1} = x_t$ otherwise
- Workers exit the labor force at the rate of ξ
- New workers enter at the rate of ξ and draw skills from a Beta distribution

Wage Determination: Hired from Unemployment



Wage Determination: $W_t \leq P_t(x, y') \leq P_t(x, y)$



$$W_t \leq P_t(x, y') \leq P_t(x, y) \quad \Rightarrow \quad \text{stay at } y, \;\; W_{t+1} = P_t(x, y')$$

Wage Determination: $P_t(x, y) < P_t(x, y')$



$$P_t(x, y') > P_t(x, y) \quad \Rightarrow \quad \text{switch to } y', \;\; W_{t+1} = P_t(x, y)$$

Wage Determination: $P_t(x, y') < W_t$



$$P_t(x, y') < W_t \quad \Rightarrow \quad \text{no counteroffer needed,} \quad W_{t+1} = W_t$$

Equilibrium

A **stochastic equilibrium** consists of:

1. A surplus function $S(x, y, z)$ satisfying the Bellman equation
2. Vacancy posting policies $v(y)$ satisfying free entry
3. Stationary distributions $(u^*(x), h^*(x, y))$ satisfying the laws of motion

Solution algorithm:

1. Compute $S(x, y, z)$ by value function iteration
2. Solve for $v(y)$, θ, λ_w, and λ_f
3. Iterate on (u_t, h_t) until convergence

Note: Computationally tractable, flows and wages only depend on current surplus

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Parameter Estimates: Simulated Method of Moments

- Model frequency: weekly
- Discount rate 5% p.a.; $\omega = 0.5$
- 14 estimated parameters, 23 moments

		Estimate			**Estimate**
Matching	α	0.8105	Production Function	a_1	0.0295
Search efficiency	s	0.0840		a_2	-0.3000
Vacancy costs	κ_0	0.2925		a_3	0.8657
	$1+\kappa_1$	1.0817		a_4	10.3074
Exogenous separation	δ	0.0078		a_5	-3.2149
Productivity shocks	σ_z	0.0472		a_6	7.6516
	ρ	0.9994	Skill accumulation	π	0.00035

$$p(x_t, y, z) = z\,(a_1 + a_2 x_t + a_3 y + a_4 x_t^2 + a_5 y^2 + a_6\, x_t y)$$

$$M(L, V) = \alpha L^{\omega} V^{1-\omega}; \quad C(v) = \frac{\kappa_0}{1+\kappa_1} v^{1+\kappa_1}$$

Data vs. Model Moments

Moment	Data	Model	Moment	Data	Model
$E[U]$	0.058	0.060	$E[sd(\log VA)]$	0.494	0.595
$E[E2U]$	0.025	0.028	$sd[sd(\log VA)]$	0.039	0.025
$E[E2E]$	0.025	0.017	Annual wage growth	0.042	0.039
$E[U2E]$	0.421	0.451			

- **Firm heterogeneity:** Identified by cross-sectional dispersion in firm value added
- **Skill accumulation:** Identified by annual wage growth (Elsby & Shapiro, 2012)

Full Table (Appendix)

Worker Types and Vacancy Creation

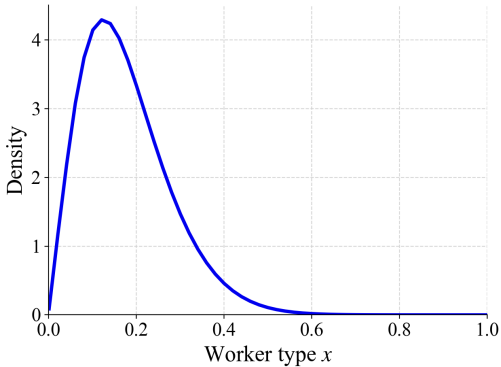


Figure: Worker Distribution

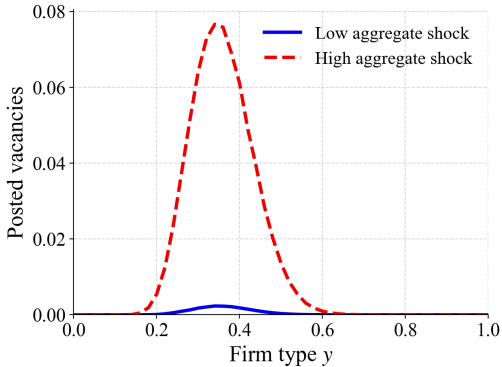
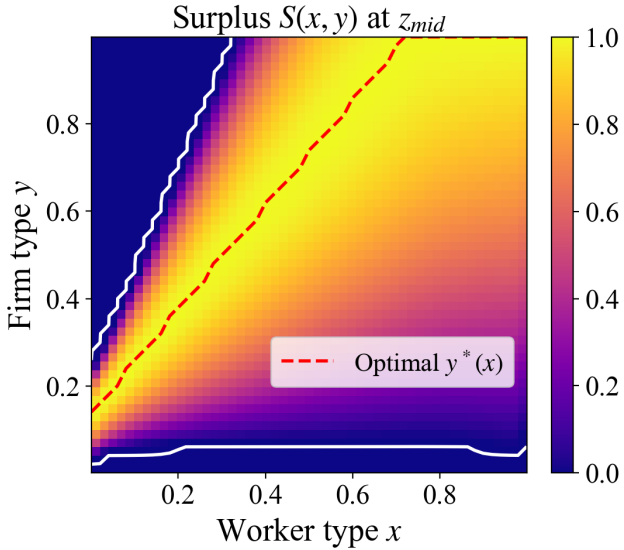


Figure: Vacancy Distribution

Match Surplus



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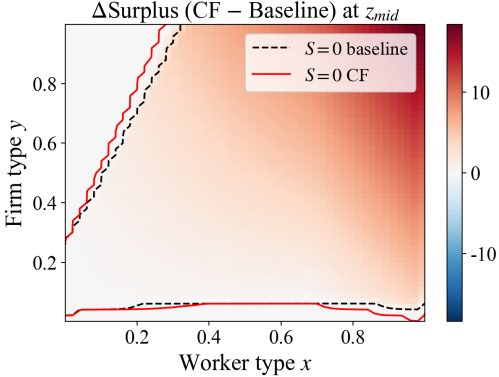
Counterfactual Exercise

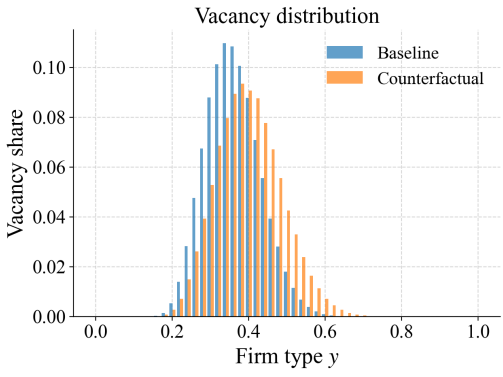
- Consider the US economy in 2012
- **Object of interest:** $\text{urate}_{\text{young}}/\text{urate}_{\text{old}}$
- **In the model:** Young = entered the labor market at most 4 years ago
- **Skill Biased Technical Change:** Rise in skill demand observed in Lightcast

	2012	2019	2024
Data	1.91	2.33	2.57
Model	1.81	1.89	1.93

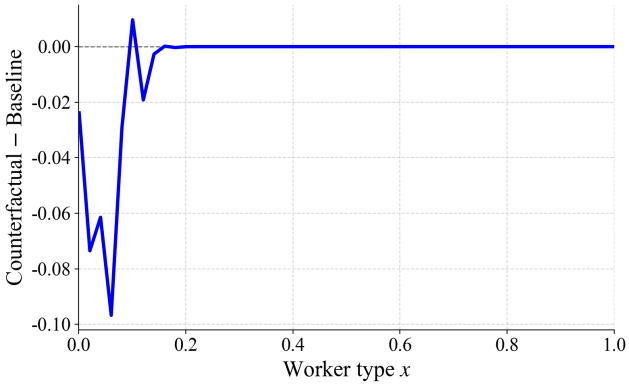
- Share explained (2019) $= \dfrac{1.89 - 1.81}{2.33 - 1.91} =$ **19%**

Counterfactual: Change in Surplus and Vacancy Creation





Consequence: Fewer Viable Vacancies For Low-Skilled Workers



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Things To Do

1. Include 2 skill dimensions in the model (soft and technical)
2. Technical skills are fixed; Soft skills are accumulated on the job
3. Derive wage moments from a simulated panel of workers

Returns to Experience

Conclusion

- Youth unemployment has been rising
- We propose a rise in soft skill demand as a potential mechanism
- Set up a search-and-matching model calibrated to the US labor market
- This explains **one-fifth** of the rise in young workers' unemployment rate

Thank You!

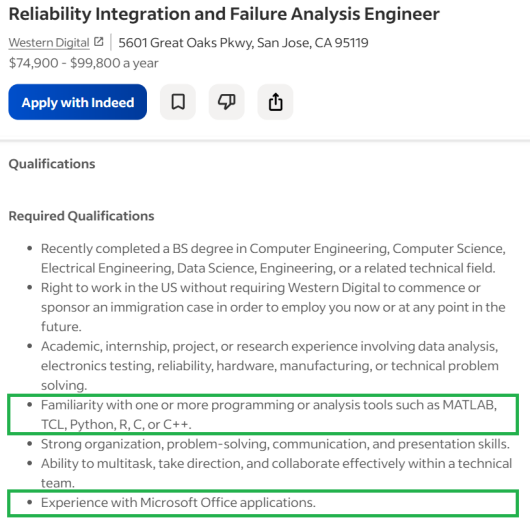
Most Common Skills in Job Postings

	Soft Skills			Technical Skills	
#	Skill	Share (%)	#	Skill	Share (%)
1	Communication	33.76	1	Writing	10.87
2	Customer Service	28.97	2	Microsoft Excel	8.48
3	Management	25.59	3	Valid Driver's License	8.33
4	Sales	19.70	4	Registered Nurse (RN)	8.15
5	Operations	16.93	5	Project Management	7.66
6	Leadership	15.63	6	Merchandising	7.43
7	Problem Solving	11.51	7	Microsoft Office	7.41
8	Detail Oriented	11.48	8	Research	7.01
9	Planning	10.16	9	Marketing	6.76
10	Interpersonal Communications	8.18	10	Computer Literacy	6.62
11	Coordinating	6.96	11	Nursing	6.61
12	Multitasking	6.83	12	Accounting	6.08
13	Verbal Communication Skills	6.78	13	Auditing	5.85
14	Organizational Skills	6.71	14	Scheduling	5.77
15	Self-Motivation	6.33	15	Finance	4.98

Share = % of Lightcast postings mentioning the skill, 2011–2024.

Back

Example: Technical Skills in a Job Posting



Reliability Integration and Failure Analysis Engineer

Western Digital | 5601 Great Oaks Pkwy, San Jose, CA 95119

$74,900 - $99,800 a year

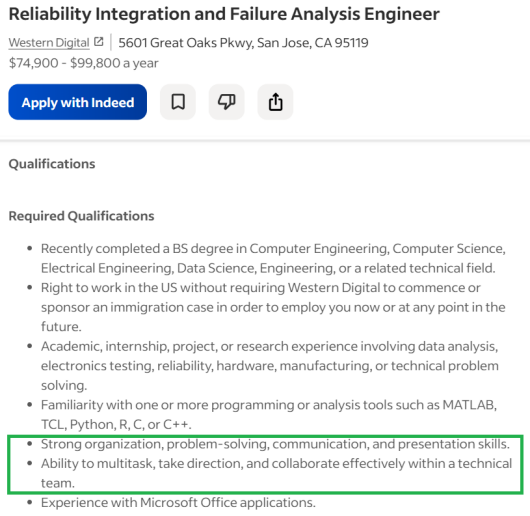
Apply with Indeed

Qualifications

Required Qualifications

- Recently completed a BS degree in Computer Engineering, Computer Science, Electrical Engineering, Data Science, Engineering, or a related technical field.
- Right to work in the US without requiring Western Digital to commence or sponsor an immigration case in order to employ you now or at any point in the future.
- Academic, internship, project, or research experience involving data analysis, electronics testing, reliability, hardware, manufacturing, or technical problem solving.
- Familiarity with one or more programming or analysis tools such as MATLAB, TCL, Python, R, C, or C++.
- Strong organization, problem-solving, communication, and presentation skills.
- Ability to multitask, take direction, and collaborate effectively within a technical team.
- Experience with Microsoft Office applications.

Example: Soft Skills in a Job Posting



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Qualifications

Required Qualifications

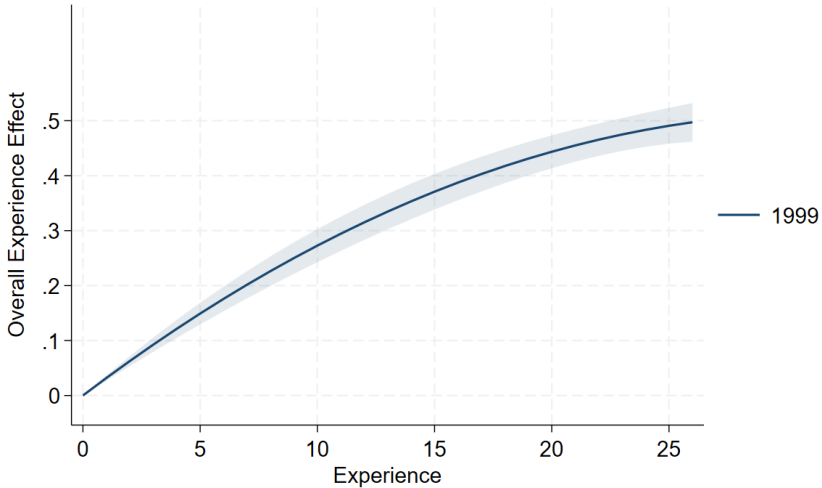
- Recently completed a BS degree in Computer Engineering, Computer Science, Electrical Engineering, Data Science, Engineering, or a related technical field.
- Right to work in the US without requiring Western Digital to commence or sponsor an immigration case in order to employ you now or at any point in the future.
- Academic, internship, project, or research experience involving data analysis, electronics testing, reliability, hardware, manufacturing, or technical problem solving.
- Familiarity with one or more programming or analysis tools such as MATLAB, TCL, Python, R, C, or C++.
- Strong organization, problem-solving, communication, and presentation skills.
- Ability to multitask, take direction, and collaborate effectively within a technical team.
- Experience with Microsoft Office applications.

Wage Regression Coefficients Over Time



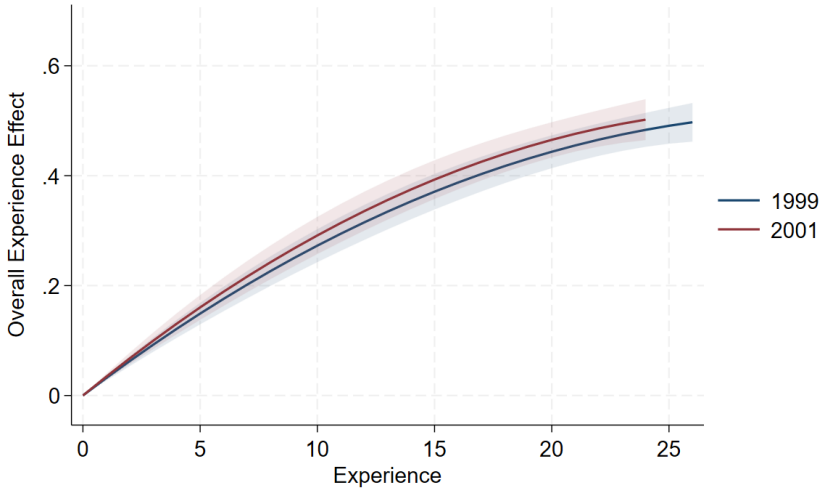
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Rising Returns To Experience



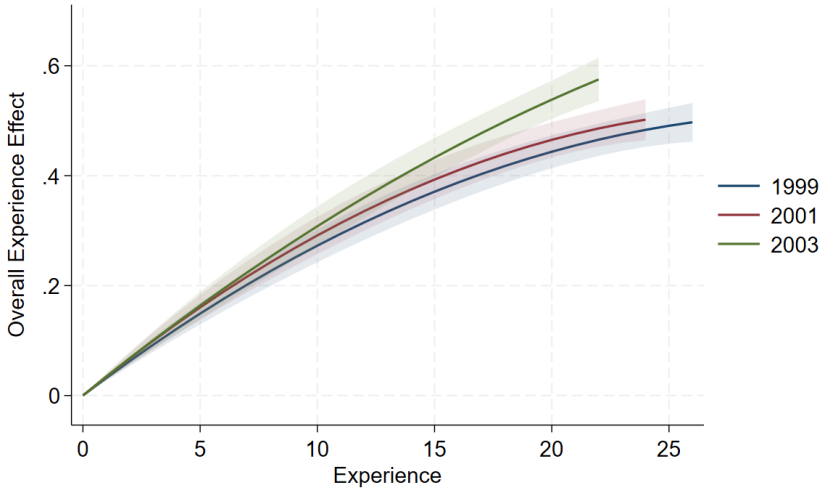
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Rising Returns To Experience



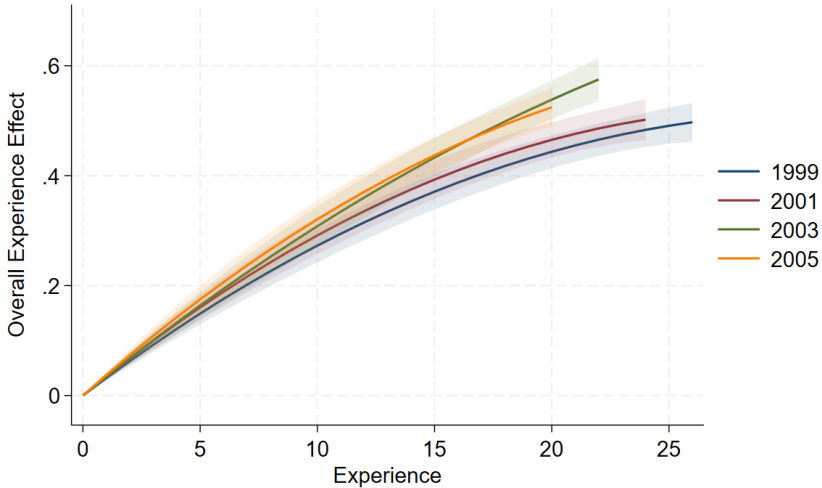
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Rising Returns To Experience



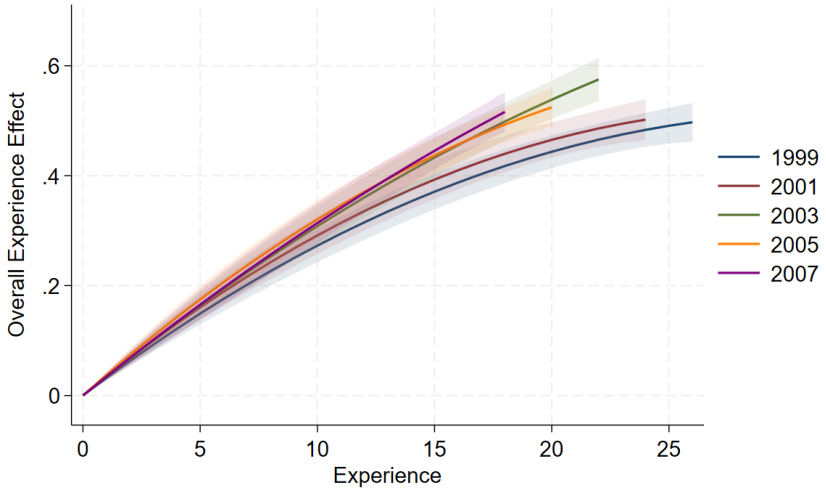
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Rising Returns To Experience



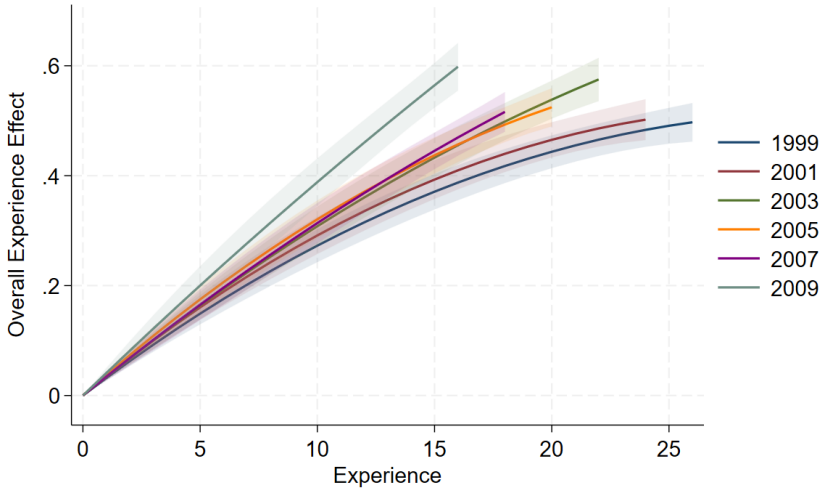
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Rising Returns To Experience



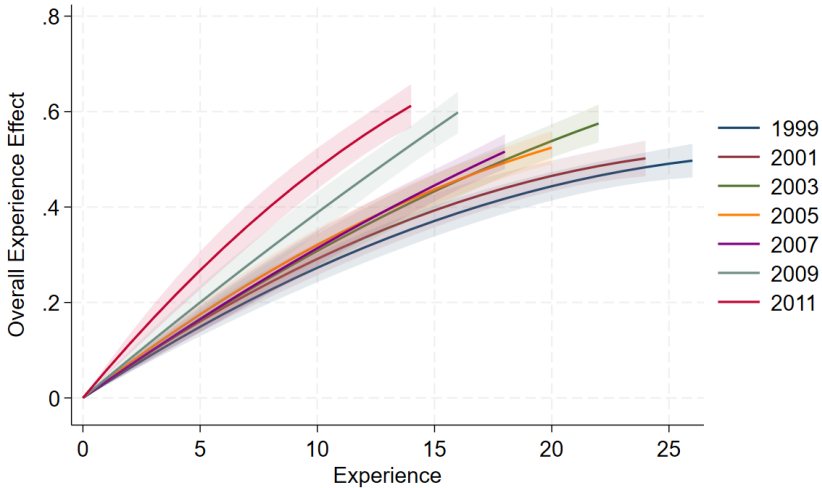
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Rising Returns To Experience



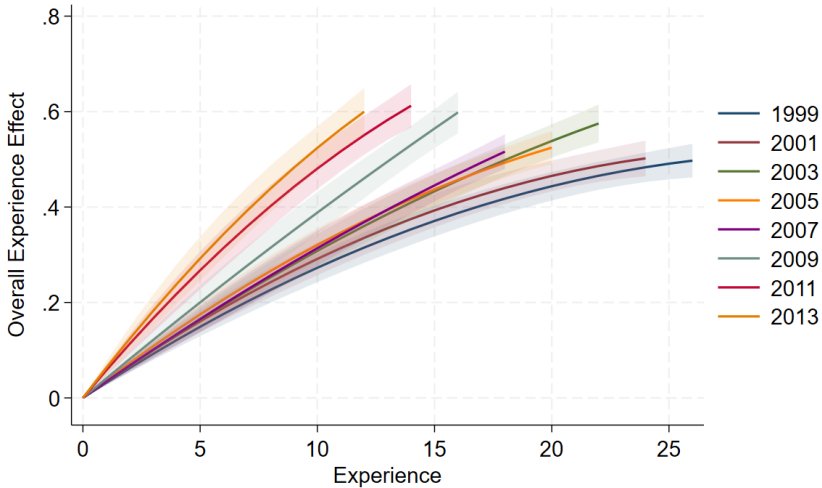
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Rising Returns To Experience



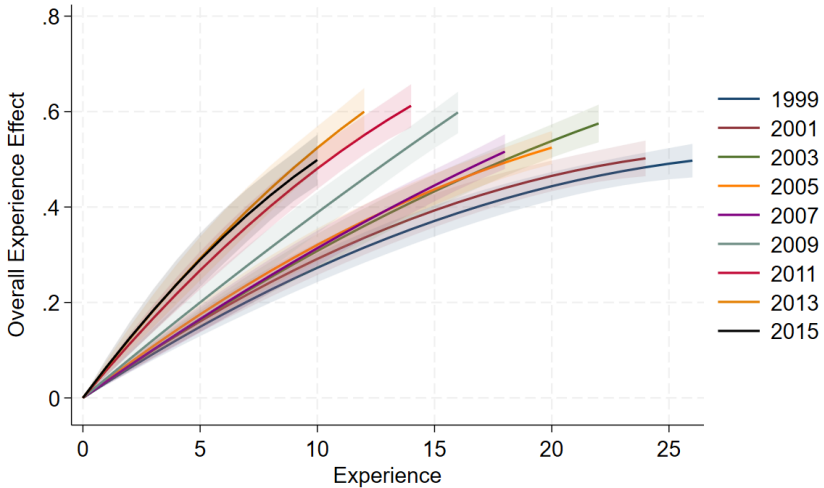
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Rising Returns To Experience



Back

Rising Returns To Experience



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Values and Surplus With Skill Accumulation

On-the-job skill accumulation: $\mathbb{E}_\pi f(x_{t+1}) \equiv \pi f(x_t + \Delta) + (1-\pi) f(x_t)$

$$B_t(x_t) = b_t(x_t) + \beta(1-\xi)\,\mathbb{E}_t\big[B_{t+1}(x_{t+1})\big]$$

$$\begin{aligned} P_t(x_t,y,z) = p_t(x_t,y,z) + \beta(1-\xi)\,\mathbb{E}_t\mathbb{E}_\pi\Big[&\big(1-(1-\delta)\mathbf{1}\{S_{t+1}(x_{t+1},y,z)\geq 0\}\big)B_{t+1}(x_{t+1}) \\ &+ (1-\delta)\mathbf{1}\{S_{t+1}(x_{t+1},y,z)\geq 0\}\,P_{t+1}(x_{t+1},y,z)\Big] \end{aligned}$$

$$\begin{aligned} S_t(x_t,y,z) = {} & p_t(x_t,y,z) - b_t(x_t) \\ & + \beta(1-\xi)(1-\delta)\,\mathbb{E}_t\mathbb{E}_\pi\Big[\mathbf{1}\{S_{t+1}(x_{t+1},y,z)\geq 0\}\,S_{t+1}(x_{t+1},y,z)\Big] \\ & + \beta(1-\xi)\,\mathbb{E}_t\big[\mathbb{E}_\pi B_{t+1}(x_{t+1}) - B_{t+1}(x_t)\big] \end{aligned}$$

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Wages With Skill Accumulation

Worker's share of surplus: $\sigma_t = \dfrac{W_t(x_t, y, z, \sigma_t) - B_t(x_t)}{P_t(x_t, y, z) - B_t(x_t)}$

$$
\begin{aligned}
w_t(x_t, y, z, \sigma_t) = {} & \sigma_t \, p_t(x_t, y, z) + (1 - \sigma_t) \, b_t(x_t) \\
& - \beta(1 - \xi)(1 - \sigma_t) \, \mathbb{E}_t \big[\mathbb{E}_\pi B_{t+1}(x_{t+1}) - B_{t+1}(x_t) \big] \\
& - \beta(1 - \xi)(1 - \delta) \, \mathbb{E}_t \mathbb{E}_\pi \Big[s \, \lambda_{w,t+1} \mathbf{1}\{ S_{t+1}(x_{t+1}, y, z) \geq 0 \} \, \Omega_{t+1}(x_{t+1}, y, z, \sigma_{t+1}) \Big]
\end{aligned}
$$

where

$$
\Omega_t(x_t, y, z, \sigma_t) = \int_{\mathcal{Y}} \Big[\min \big\{ S_t(x_t, y, z), \, \max\{ S_t(x_t, y', z), \, \sigma_t S_t(x_t, y, z) \} \big\} - \sigma_t S_t(x_t, y, z) \Big] \frac{v_t(y')}{V_t} \, dy'
$$

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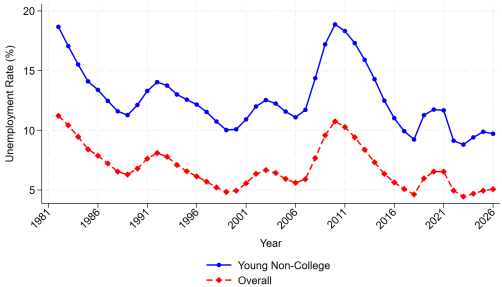
Data vs. Model Moments (Full)

Moment	Data	Model	Moment	Data	Model
$E[U]$	0.058	0.060	$\text{sd}[U]$	0.191	0.175
$E[UE]$	0.421	0.451	$\text{sd}[UE]$	0.127	0.131
$E[EU]$	0.025	0.028	$\text{sd}[EU]$	0.100	0.069
$E[EE]$	0.025	0.017	$\text{sd}[EE]$	0.095	0.105
$E[V/U]$	0.634	0.091	$\text{sd}[V/U]$	0.381	0.355
E[sd labor prod]	0.494	0.595	sd[sd labor prod]	0.039	0.025
$\text{sd}[V]$	0.206	0.185	$\text{corr}[V, VA]$	0.721	0.981
$\text{sd}[VA]$	0.033	0.034	$\text{corr}[UE, VA]$	0.878	0.951
$\text{autocorr}[VA]$	0.931	0.985	$\text{corr}[EU, VA]$	−0.717	−0.773
$\text{corr}[V, U]$	−0.846	−0.951	$\text{corr}[UE, EE]$	0.695	0.976
$\text{corr}[U, VA]$	−0.860	−0.951	corr[sd labor prod, ΔVA]	−0.366	−0.318
Annual wage growth	0.042	0.039			

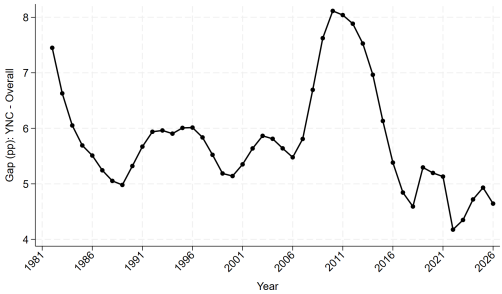
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Unemployment Rate of Young Non-College Workers

Young Non-College vs. Overall



Gap: Young Non-College − Overall



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Counterfactual Exercise

- Consider the US economy in 2012
- **Object of interest:** $\text{urate}_{\text{young}}/\text{urate}_{\text{overall}}$
- **In the model:** Bottom 10% of worker types = young
- **Skill Biased Technical Change:** Rise in skill demand observed in Lightcast

	2012	2019	2024
Data	1.77	2.09	2.26
Model	1.68	1.74	1.77

- Share explained (2019) $= \dfrac{1.74 - 1.68}{2.09 - 1.77} =$ **19%**